

SASI DEGREE COLLEGE
VELIVENNU-534329
(AFFILATED TO ADIKAVI NANNAYA UNIVERSITY)



HEALTH INSURANCE
SUBMITTED TO
SASI DEGREE COLLEGE
BY STUDENTS OF BSC
IN PARTIAL FULFILLMENT OF
UNDERGRADUATION PROGRAM

2023-2026

SASI DEGREE COLLEGE
(AFFILATED TO ADIKAVI NANNAYA
UNIVERSITY, CODE:-401)
VELIVENNU



HEALTH INSURANCE



BY

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INTERNSHIP COMPLETION LETTER

This is to certify that **SUSHRUTAV(234017170043)** undergoing degree in **B.SC COMPUTER SCIENCE** at **SASI DEGREE COLLEGE, VELIVENNU** have successfully completed her internship in **Sri Tulasi Edtech Pvt Ltd** as per enclosed details.

Internship summary:

- **Duration:**One Semester
- **Time period:**15-12-2025 to 29-03-2026
- **Project title:** Health Insurance
- **Project type:**Group Project
- **Internship status:**Completed

Candidate internship performance is satisfactory as per the initially stated requirements and we wish her all the best for her future endeavors.

For Sri Tulasi Edtech Pvt Ltd

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HumanResourcesDepartment

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INTERNSHIP COMPLETION LETTER

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Internship summary:

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- **Time period:**15-12-2025 to 29-03-2026
- **Project title:** Health Insurance
- **Project type:**Group Project
- **Internship status:**Completed

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Internship summary:

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- **Time period:**15-12-2025 to 29-03-2026
- **Project title:** Health Insurance
- **Project type:**Group Project
- **Internship status:**Completed

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- **Time period:** 15-12-2025 to 29-03-2026
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- **Time period:** 15-12-2025 to 29-03-2026
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- **Time period:**15-12-2025 to 29-03-2026
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This is to certify that **PILLI AMRUTHA VARSHINI (234017170037)** undergoing degree in **B.SC COMPUTER SCIENCE** at **SASI DEGREE COLLEGE,VELIVENNU** have successfully completed her internship in **Sri Tulasi Edtech Pvt Ltd** as per enclosed details.

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This is to certify that **PINISETTI LAKSHMI DURGA(234017170038)** undergoing degree in **B.SC COMPUTER SCIENCE** at **SASI DEGREE COLLEGE,VELIVENNU** have successfully completed her internship in **Sri Tulasi Edtech Pvt Ltd** as per enclosed details.

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This is to certify that **SHAIK JESSY(234017170040)** undergoing degree in **B.SC COMPUTERSCIENCE** at **SASI DEGREE COLLEGE,VELIVENNU** have successfully completed her internship in **Sri Tulasi Edtech Pvt Ltd** as per enclosed details.

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INTERNSHIP COMPLETION LETTER

This is to certify that **SIRRA NAVEEN KUMAR(234017170042)**undergoing degree in **B.SC COMPUTER SCIENCE** at **SASI DEGREE COLLEGE,VELIVENNU** have successfully completed his internship in **Sri Tulasi Edtech Pvt Ltd** as per enclosed details.

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- **Time period:**15-12-2025 to 29-03-2026
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For SriTulasiEdtechPvtLtd

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INTERNSHIP COMPLETION LETTER

This is to certify that **THADI SHAHAMTHI(234017170044)** undergoing degree in **B.SC COMPUTER SCIENCE** at **SASI DEGREE COLLEGE,VELIVENNU** have successfully completed her internship in **Sri Tulasi Edtech Pvt Ltd** as per enclosed details.

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- **Project type:**GroupProject
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INTERNSHIP COMPLETION LETTER

This is to certify that **VEERAMALLU MOUNIKA (234017170045)** undergoing degree in **B.SC COMPUTER SCIENCE** at **SASI DEGREE COLLEGE,VELIVENNU** have successfully completed her internship in **Sri Tulasi Edtech Pvt Ltd** as per enclosed details.

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For SriTulasiEdtechPvtLtd

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This is to certify that **VEERAMALLU SUJATHA(234017170046)** undergoing degree in **B.SC COMPUTER SCIENCE** at **SASI DEGREE COLLEGE,VELIVENNU** have successfully completed her internship in **Sri Tulasi Edtech Pvt Ltd** as per enclosed details.

Internship summary:

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- **Time period:**15-12-2025 to 29-03-2026
- **Project title:** Health Insurance
- **Project type:**Group Project
- **Internshipstatus:**Completed

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For SriTulasiEdtechPvtLtd

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INTERNSHIP COMPLETION LETTER

This is to certify that **YARRA BHAVAN SAI PRAVEEN KUMAR (234017170047)** undergoing degree in **B.SC COMPUTERSCIENCE** at **SASI DEGREE COLLEGE, VELIVENNU** have successfully completed his internship in **Sri Tulasi Edtech Pvt Ltd** as per enclosed details.

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- **Internship status:** Completed

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For SriTulasiEdtechPvtLtd

Authorized Signature

Health Insurance Internship Report

Submitted in accordance with the requirement for the degree of Bsc(CS),3rd year,6th sem internship

Name of the College : Sasi Degree College

Name of the Faculty Guide :G Satya Prakash

Department :Computer Science

Duration of the Internship :3months

Internship Starting Date :15thDecember2025

Internship Ending Date :29thMarch 2026

No. of Working Days :64 Days

Program of Study :Insurance

Year of Study :2026 3rdYear

AcademicYear :2025-2026

Date of Submission : 25th March2026

Acknowledgment

We have taken efforts in this project; however, it would not have been possible without the kind Support and help of many individuals and organization. We would like to extend our sincere thanks to all of them.

We are also thankful to **Smt. Karuturi Radhika Garu, M.Sc.** honorable Principal of Sasi Degree College, Velivenu and I express my sincere gratitude and deep sense of respect for making me available with all the required assistance and for her support and inspiration to carry out this thesis in the Institute.

We are also thankful to the coordinator **Sri. G.Satya Prakash, MTech** for his encouragement and cooperation to complete this thesis in time.

We are highly to our guide **Sri. V R K Pavan Kumar, MCA** for his guidance and constant supervision as Well as for providing necessary information regarding the project & also for the support in completing the project. I would like to express my gratitude towards our parents and members of “**SRI TULASI EDTECH PVT LTD**” for their kind cooperation and encouragement which helped me in completion of this project. I am thankful to and fortunate enough to get constant encouragement, support and guidance from all SRI TULASI EDTECH TEAM.

SELF-ASSESSMENT

We are here to complete our project: Health Insurance. In this project we developed a website which is useful for students.

One of the most important skills developed during this project was the ability to use Python for data analysis and visualization. Python played a key role in handling the dataset, performing calculations, and generating graphical representations of the data. Libraries such as Pandas, NumPy, Matplotlib, and Seaborn were used to process the dataset and create visualizations like bar charts, distribution plots, and heatmaps. These tools helped transform raw data into meaningful visual insights.

Through this project, I also improved my understanding of the complete data analysis workflow. The project followed several structured steps including problem definition, dataset exploration, data cleaning, univariate analysis, bivariate analysis, correlation analysis, and result interpretation. Each stage helped in building a deeper understanding of the relationships between different variables such as gender, parental education level, test preparation courses, and student exam scores.

Student's Signature

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ABSTRACT

Health insurance plays a crucial role in modern healthcare systems by helping individuals manage medical expenses. Insurance companies determine the premium amount based on various factors such as age, body mass index (BMI), smoking habits, and geographic location.

This project focuses on analyzing a health insurance dataset using Python-based data analysis techniques. The dataset contains information about individuals including their age, gender, BMI, number of children, smoking status, region, and medical insurance charges.

The primary objective of this project is to perform Exploratory Data Analysis (EDA) in order to understand how different factors influence insurance costs.

Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn are used to perform data cleaning, analysis, and visualization.

Through data analysis, the project identifies patterns and relationships between variables such as:

- age and insurance charges
- BMI and medical costs
- smoking status and insurance premiums
- regional differences in insurance charges

The findings provide valuable insights into the factors affecting health insurance costs and demonstrate the practical application of data analysis techniques using Python. [insurance.csv](#)

1.INTRODUCTION

Health insurance is an important financial tool that protects individuals from high medical expenses. Insurance companies calculate the premium amount by analyzing several factors related to an individual's health and lifestyle.

In recent years, data analysis has become an essential tool in the insurance industry. Insurance companies use large datasets to analyze risk factors and determine appropriate premium charges.

Data analysis allows insurance providers to:

- understand customer characteristics
- identify risk patterns
- estimate healthcare costs
- optimize pricing strategies

The insurance dataset used in this project provides information about individuals and their insurance charges.

By analyzing this dataset, we can identify relationships between demographic factors and medical costs.

1.1 Importance of Data Analysis in Insurance

Data analysis plays a significant role in the insurance industry.

Insurance companies collect large amounts of data related to customers' health, lifestyle, and demographics. Analyzing this data helps companies estimate the probability of medical expenses and determine appropriate insurance premiums.

Benefits of data analysis in insurance include:

- risk assessment
- fraud detection
- cost prediction
- policy optimization

Using data analysis techniques, companies can make more informed decisions and improve their services.

1.2 Objectives of the Project

The objectives of this project include:

1. To analyze the health insurance dataset.
2. To understand the relationship between personal attributes and insurance charges.
3. To perform data cleaning and preprocessing.
4. To separate categorical and numerical variables.
5. To perform exploratory data analysis.
6. To visualize relationships between variables.
7. To identify key factors affecting insurance costs.

2. DATASET DESCRIPTION

The dataset used in this project contains information about **health insurance customers**.

Dataset characteristics:

- **Rows:** 1338

- **Columns:** 7

Each row represents an individual and each column represents a feature describing that individual.

2.1 Dataset Features

The dataset contains the following columns:

Column	Description
--------	-------------

age	Age of the individual
-----	-----------------------

sex	Gender of the individual
-----	--------------------------

bmi	Body Mass Index
-----	-----------------

children	Number of children covered by insurance
----------	---

smoker	Whether the individual is a smoker
--------	------------------------------------

region	Residential region
--------	--------------------

charges	Medical insurance charges
---------	---------------------------

3. DATA TYPES

The dataset contains both categorical and numerical variables.

Categorical Variables

Categorical variables contain text-based categories.

Column

sex

smoker

region

These columns describe qualitative attributes of individuals.

Numerical Variables

Numerical variables contain numeric values used for statistical analysis.

Column

age

bmi

children

charges

These columns are used to perform statistical analysis and identify relationships between variables.

4. DATA PREPROCESSING

Data preprocessing is an essential step before performing data analysis.

It includes:

- handling missing values
- removing duplicates

- converting data types
- identifying outliers

4.1 Checking Missing Values

Example code:

```
df.isnull().sum()
```

The dataset does not contain missing values, which means it is clean and suitable for analysis.

4.2 Checking Duplicate Records

Duplicate records can affect analysis results.

Example code:

```
df.duplicated().sum()
```

If duplicates exist, they can be removed using:

```
df.drop_duplicates()
```

5.EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis helps understand patterns and relationships in the dataset.

EDA includes:

- univariate analysis
- bivariate analysis
- correlation analysis

6.UNIVARIATE ANALYSIS

Univariate analysis studies one variable at a time.

Age Distribution

A histogram is used to visualize the distribution of age.

Example code:

```
sns.histplot(df['age'])
```

This helps understand the age distribution of insured individuals.

BMI Distribution

BMI distribution helps identify the health characteristics of individuals.

Higher BMI values may indicate obesity, which can affect insurance charges.

Insurance Charges Distribution

Analyzing insurance charges helps understand how costs vary across individuals.

7.BIVARIATE ANALYSIS

Bivariate analysis studies relationships between two variables.

Age vs Insurance Charges

Older individuals may have higher medical expenses, which can increase insurance costs.

Visualization:

Scatter plot

BMI vs Insurance Charges

Higher BMI values may increase the risk of health problems, which may lead to higher insurance costs.

Smoker vs Charges

Smokers generally have higher insurance charges due to increased health risks.

8.CATEGORICAL VARIABLE ANALYSIS

Sex Distribution

Bar charts can be used to visualize the number of male and female individuals.

Example:

```
sns.countplot(x='sex', data=df)
```

Smoker Distribution

This visualization shows how many individuals in the dataset are smokers. Smoking significantly affects insurance premiums.

Region Distribution

This shows how individuals are distributed across different geographic regions.

9.CORRELATION ANALYSIS

Correlation analysis helps identify relationships between numerical variables.

Example:

```
sns.heatmap(df.corr(), annot=True)
```

The heatmap shows relationships between:

- age
- bmi
- children
- charges

10.KEY INSIGHTS

From the analysis, several insights can be observed:

- Smokers have significantly higher insurance charges.
- Age has a positive correlation with insurance costs.
- Higher BMI values are associated with increased medical expenses.
- Regional differences exist in insurance costs.

11.LIMITATIONS

The project has certain limitations:

- dataset size is limited
- dataset contains limited features
- lifestyle factors beyond smoking are not included

12.FUTURE WORK

Future improvements may include:

- building machine learning models to predict insurance charges
- using larger datasets

- including additional health indicators

13.DATA VISUALIZATION AND GRAPH INTERPRETATION

Data visualization plays an important role in exploratory data analysis because it helps transform raw data into meaningful graphical representations. Instead of examining numbers in tables, visualizations allow analysts to easily identify patterns, trends, and relationships between variables.

In this project, several types of visualizations were used to analyze the insurance dataset. These visualizations include bar charts, histograms, box plots, scatter plots, and heatmaps.

Visualization tools such as Matplotlib and Seaborn were used to generate these graphs.

13.1 Age Distribution Analysis

The age distribution graph shows how individuals are distributed across different age groups in the dataset.

Histogram visualization helps identify whether the data is evenly distributed or concentrated in certain age ranges.

Python Code Example

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
sns.histplot(df['age'], bins=20)
plt.title("Age Distribution")
plt.show()
```

Interpretation:

The graph indicates that the dataset contains individuals from different age groups ranging from young adults to older individuals. Most individuals fall within the middle age range.

This suggests that health insurance is commonly purchased by people across a wide range of age groups.

13.2 BMI Distribution

BMI (Body Mass Index) is an important health indicator used to measure body fat based on height and weight.

The BMI distribution helps understand the health profile of individuals in the dataset.

Python Code Example

```
sns.histplot(df['bmi'], bins=20)
plt.title("BMI Distribution")
plt.show()
```

Interpretation:

The BMI distribution shows that most individuals fall within the normal to overweight range. Extremely high BMI values are less common.

Higher BMI values may indicate increased health risks, which may result in higher insurance charges.

13.3 Gender Distribution

Gender distribution shows the number of male and female individuals present in the dataset.

Python Code Example

```
sns.countplot(x='sex', data=df)
plt.title("Gender Distribution")
plt.show()
```

Interpretation:

The dataset contains both male and female individuals. The distribution appears relatively balanced, meaning both genders are well represented in the dataset.

This balance ensures that the dataset provides a fair representation of both genders for analysis.

13.4 Smoking Status Analysis

Smoking status is one of the most important factors affecting health insurance costs.

Python Code Example

```
sns.countplot(x='smoker', data=df)
plt.title("Smoker vs Non-Smoker Distribution")
plt.show()
```

Interpretation:

The graph shows that most individuals in the dataset are non-smokers. However, smokers tend to have significantly higher insurance costs due to increased health risks.

13.5 Region Distribution

The region column represents the geographic location of individuals.

Python Code Example

```
sns.countplot(x='region', data=df)
plt.title("Region Distribution")
plt.show()
```

Interpretation:

The dataset includes individuals from four regions:

- Northeast
- Northwest
- Southeast
- Southwest

The distribution across regions appears relatively balanced.

14.RELATIONSHIP ANALYSIS BETWEEN VARIABLES

Understanding relationships between variables helps identify the factors that influence insurance charges.

14.1 Age vs Insurance Charges

Scatter plots are used to visualize relationships between two numerical variables.

Python Code Example

```
sns.scatterplot(x='age', y='charges', data=df)
plt.title("Age vs Insurance Charges")
plt.show()
```

Interpretation:

The graph shows that insurance charges generally increase with age. Older individuals tend to have higher medical expenses, which results in higher insurance premiums.

14.2 BMI vs Insurance Charges

Python Code Example

```
sns.scatterplot(x='bmi', y='charges', data=df)
plt.title("BMI vs Insurance Charges")
plt.show()
```

Interpretation:

The relationship between BMI and insurance charges indicates that individuals with higher BMI values often have higher medical costs. However, BMI alone does not completely determine insurance charges because other factors also influence the cost.

14.3 Smoking Status vs Insurance Charges

Python Code Example

```
sns.boxplot(x='smoker', y='charges', data=df)
plt.title("Smoking Status vs Insurance Charges")
plt.show()
```

Interpretation:

The boxplot clearly shows that smokers have significantly higher insurance charges compared to non-smokers.

This occurs because smoking increases the risk of serious health conditions such as lung disease, heart disease, and cancer.

15.CORRELATION ANALYSIS

Correlation analysis helps measure the strength of relationships between numerical variables.

Correlation values range between:

- -1 (strong negative correlation)
- 0 (no correlation)
- +1 (strong positive correlation)

Python Code Example

```
sns.heatmap(df.corr(), annot=True)
plt.title("Correlation Heatmap")
plt.show()
```

Interpretation:

The heatmap shows the correlation between:

- age
- bmi
- children

- charges

Key observations include:

- Age has a positive correlation with insurance charges.
- BMI also shows some correlation with charges.
- Number of children has little impact on insurance costs.

16. STATISTICAL SUMMARY

Statistical analysis helps understand the basic properties of the dataset.

Python Code Example

```
df.describe()
```

This provides the following information:

- Mean
- Standard deviation
- Minimum value
- Maximum value
- Quartiles

Statistical summary helps identify the range and distribution of numerical variables.

17.KEY FINDINGS

After performing exploratory data analysis, several important insights were identified.

Key findings include:

1. Smoking is the most significant factor influencing insurance charges.
2. Older individuals generally pay higher insurance premiums.
3. Individuals with higher BMI values may have higher medical expenses.
4. The number of children does not significantly affect insurance charges.
5. Regional differences in insurance charges are relatively small.

18.PROJECT WORKFLOW

The following workflow was followed during the project.

Start



Load Insurance Dataset



Understand Dataset Structure



Check Missing Values



Separate Categorical and Numerical Columns



Perform Data Cleaning



Exploratory Data Analysis



Data Visualization



Correlation Analysis



Interpret Results



Generate Insights



End

This workflow ensures a systematic approach to data analysis.

19.CONCLUSION

The health insurance dataset analysis project demonstrates how data analysis techniques can be applied to real-world healthcare data.

Using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn, the dataset was analyzed to identify patterns and relationships between personal attributes and insurance charges.

The analysis revealed that smoking status, age, and BMI are important factors influencing health insurance costs.

This project highlights the value of data analysis in helping insurance companies understand risk factors and determine appropriate insurance premiums.

This project demonstrates how data analysis techniques can be used to analyze health insurance datasets.

Using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn, the dataset was analyzed to identify relationships between personal attributes and insurance charges.

The analysis highlights the importance of lifestyle factors such as smoking and BMI in determining healthcare costs.

This project provides practical experience in applying data analysis techniques to real-world datasets.

20.PROGRAMMING

```
import pandas as pd
from pyodide.http import open_url url="https://
raw.githubusercontent.com/AvinashMelam/visa_csv/main/
insurance.csv"
df=pd.read_csv(open_url(url))
df.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

df.types()

```
df.dtypes
```

```
age          int64
sex          object
bmi          float64
children     int64
smoker       object
region       object
charges      float64
dtype: object
```

df.shape

(1338, 7)

df.tail

```

: <bound method NDFrame.tail of
region      charges
0         19  female  27.900
1         18   male  33.770
2         28   male  33.000
3         33   male  22.705
4         32   male  28.880
...      ...      ...      ...
1333      50   male  30.970
1334      18  female  31.920
1335      18  female  36.850
1336      21  female  25.800
1337      61  female  29.070

age    sex    bmi  children  smo
0     yes southwest  16884.92400
1     no  southeast   1725.55230
2     no  southeast   4449.46200
3     no  northwest  21984.47061
4     no  northwest   3866.85520
...    ...      ...      ...
3     no  northwest  10600.54830
0     no  northeast   2205.98080
0     no  southeast   1629.83350
0     no  southwest   2007.94500
0    yes  northwest  29141.36030

[1338 rows x 7 columns]>

```

```
df.size
```

```
9366
```

```
df.columns
```

```
Index(['age', 'sex', 'bmi', 'children', 'smoker', 'region', 'charges'], dtype
e='object')
```

```
len(df.columns)
```

```
7
```

```
df["sex"].dtype
```

```
dtype('O')
```

```
df["sex"].unique()
```

```
array(['female', 'male'], dtype=object)
```

```
df["region"].unique()
```

```
array(['southwest', 'southeast', 'northwest', 'northeast'], dtype=object)
```

```
len(df["region"].unique())
```

```
4
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
l_con_count=[]
```

```
for i in df["region"].unique():
```

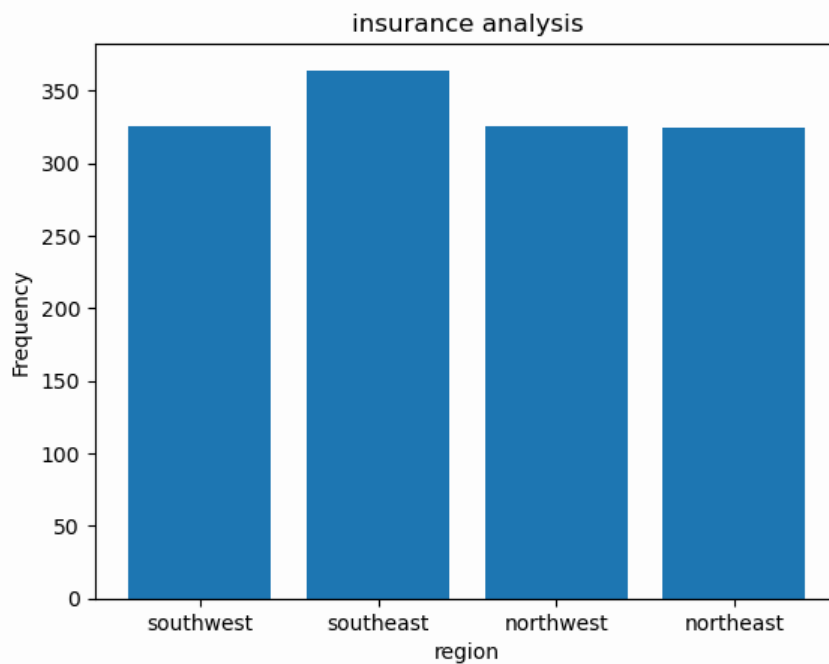
```
    con=df["region"]==i
```

```
    l_con_count.append(len(df[con]))
```

```
l_count=list(df["region"].unique())
print(l_count) print(l_con_count
['southwest', 'southeast', 'northwest', 'northeast'] [325, 364, 325, 324]
d={"region":l_count,
  "freq":l_con_count}
D
{'region': ['southwest', 'southeast', 'northwest', 'northeast'], 'freq': [325, 364,
325, 324]}
df_cont=pd.DataFrame(d,index=list(range(1,5)))
df_cont
```

	region	freq
1	southwest	325
2	southeast	364
3	northwest	325
4	northeast	324

```
plt.bar(l_count,l_con_count)
plt.xlabel("region")
plt.ylabel("Frequency")
plt.title("insurance analysis")
```



```
df_cat=df.select_dtypes(include="object")
df_cat
```

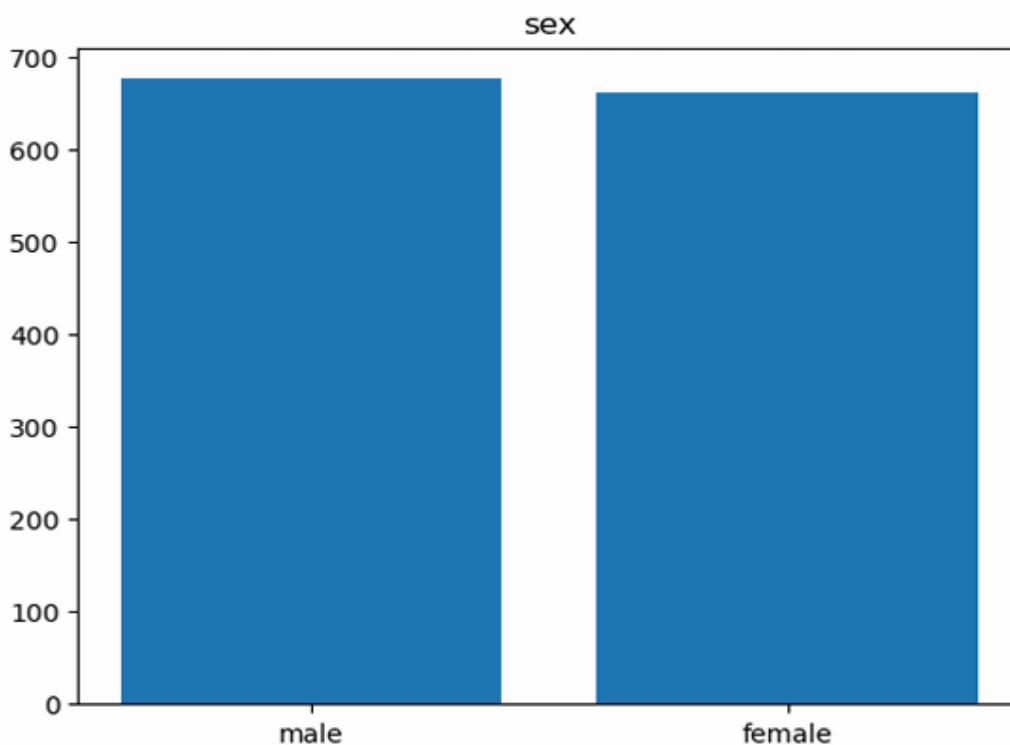
```
0] :
      sex  smoker  region
0  female    yes southwest
1   male     no  southeast
2   male     no  southeast
3   male     no northwest
4   male     no northwest
...     ...     ...     ...
1333  male     no northwest
1334 female     no  northeast
1335 female     no  southeast
1336 female     no southwest
1337 female    yes northwest

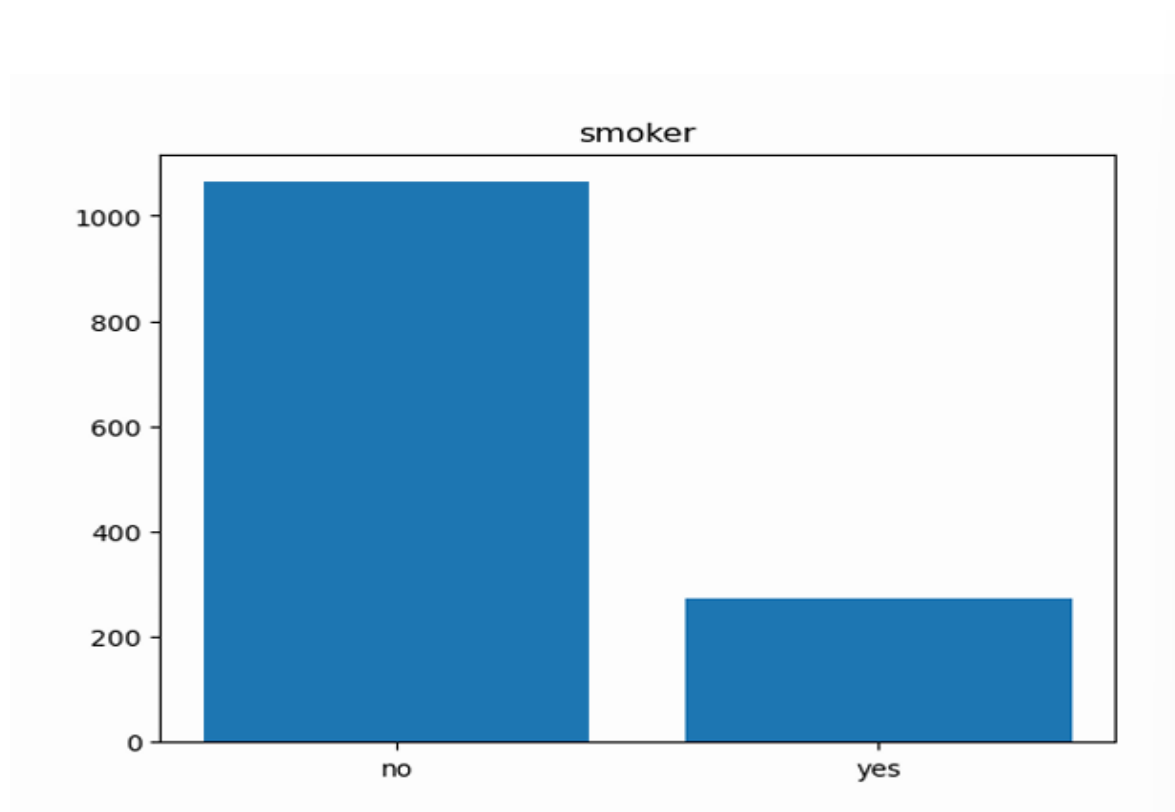
1338 rows x 3 columns
```

```
df["region"].value_counts()
```

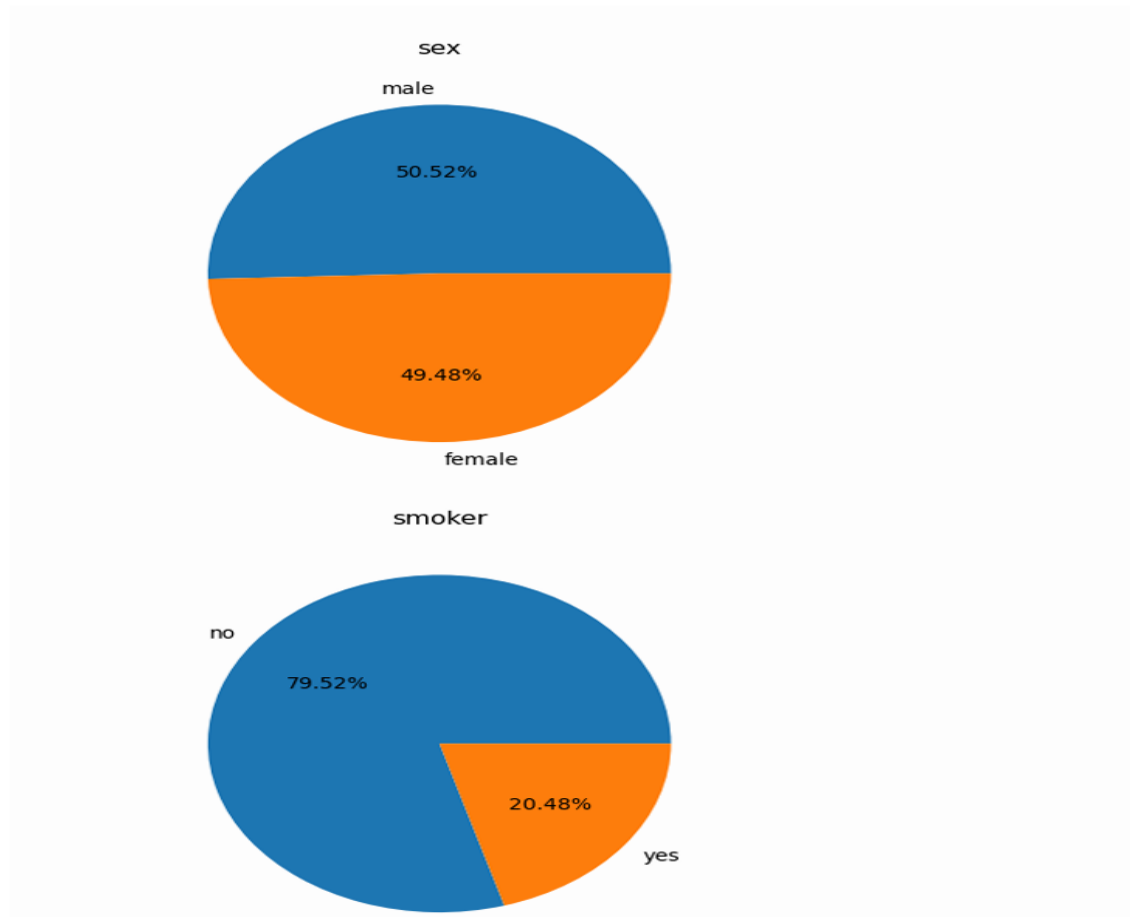
```
region
southeast 364
southwest 325
northwest 325
northeast 324
Name: count, dtype: int64
df["region"].value_counts().index
Index(['southeast', 'southwest', 'northwest', 'northeast'], dtype='object', name='region')
df["region"].value_counts().values
array([364, 325, 325, 324], dtype=int64)

for i in df_cat.columns:
    cat=df_cat[i].value_counts().index
    freq=df_cat[i].value_counts().values
    plt.bar(cat,freq)
    plt.title(i)
    plt.show()
```





```
for i in df_cat.columns:  
    cat=df_cat[i].value_counts().index  
    freq=df_cat[i].value_counts().values  
    plt.pie(freq,labels=cat,autopct="%.2f%%")  
    plt.title(i)  
    plt.savefig(f"{i}_pie.jpg")  
    plt.show()
```

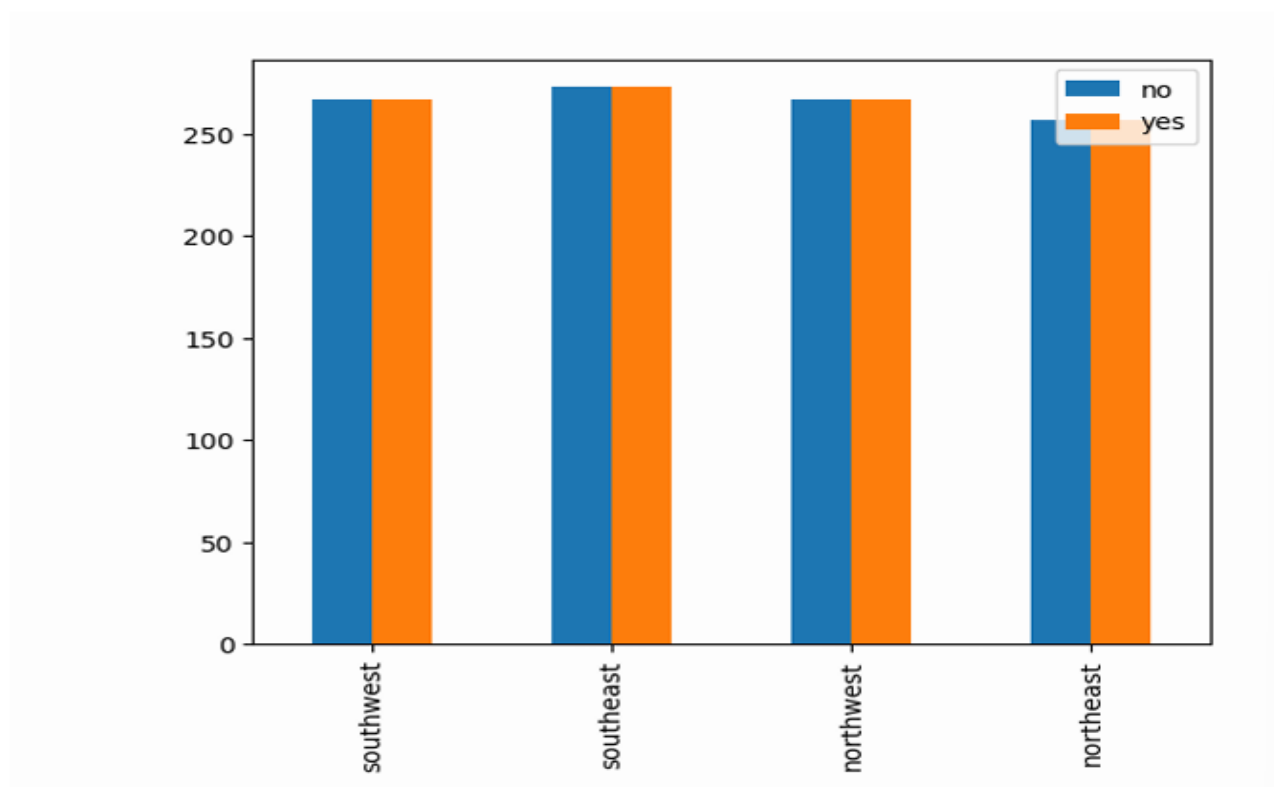


```
conA=df["region"]=="southwest"
conB=df["smoker"]=="no"
conC=df["smoker"]=="yes" print(len(df[conA & conB]))
print(len(df[conA & conC]))
267 58
```

```
cont_cert=[]
cont_deni=[]
for i in df["region"].unique():
    conA=df["region"]==i
    conB=df["smoker"]=="no"
    conC=df["smoker"]=="yes"
    cont_cert.append(len(df[conA & conB]))
    cont_deni.append(len(df[conA & conC]))
    cont=list(df["region"].unique())
print(cont)
print(cont_cert)
print(cont_deni)
['southwest', 'southeast', 'northwest', 'northeast'] [267, 273, 267, 257] [267,
273, 267, 257]
d={
    "no":cont_cert,
    "yes":cont_deni
}
df_cont_con=pd.DataFrame(d,index=cont)
df_cont_con
```

	no	yes
southwest	267	267
southeast	273	273
northwest	267	267
northeast	257	257

```
df_cont_con.plot(kind="bar")
```



```
df_num=df.select_dtypes(exclude="object")
```

```
df_num
```

```
df_num["charges"]
```

```
: 0      16884.92400
  1      1725.55230
  2      4449.46200
  3     21984.47061
  4      3866.85520
    ...
1333    10600.54830
1334     2205.98080
1335     1629.83350
1336     2007.94500
1337    29141.36030
Name: charges, Length: 1338, dtype: float64
```

```
import numpy as np
np.min(df_num["charges"])
np.float64(1121.8739)
np.mean(df_num["charges"])
np.float64(13270.422265141257)
np.median(df_num["charges"])
np.float64(9382.033)
np.max(df_num["charges"])
np.float64(63770.42801)
np.percentile(df_num["charges"],25)
np.float64(4740.28715)
np.count_nonzero(df_num["charges"])
1338
df_num.describe()
```

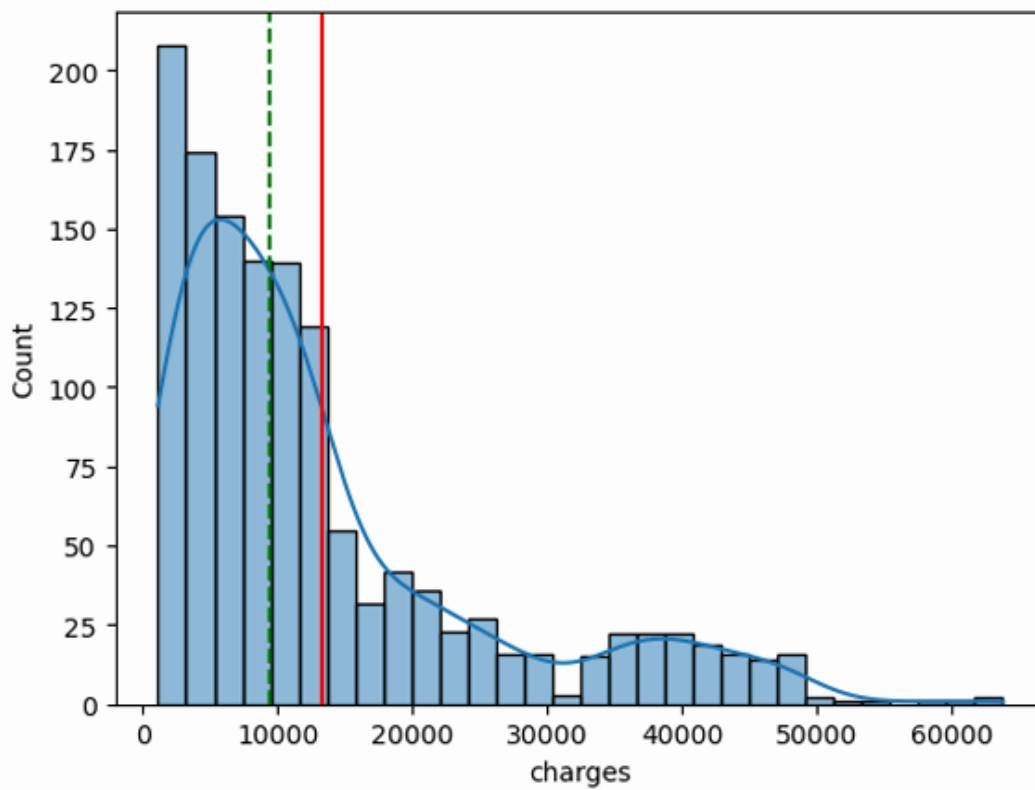
82]:

	age	bmi	children	charges
count	1338.000000	1338.000000	1338.000000	1338.000000
mean	39.207025	30.663397	1.094918	13270.422265
std	14.049960	6.098187	1.205493	12110.011237
min	18.000000	15.960000	0.000000	1121.873900
25%	27.000000	26.296250	0.000000	4740.287150
50%	39.000000	30.400000	1.000000	9382.033000
75%	51.000000	34.693750	2.000000	16639.912515
max	64.000000	53.130000	5.000000	63770.428010

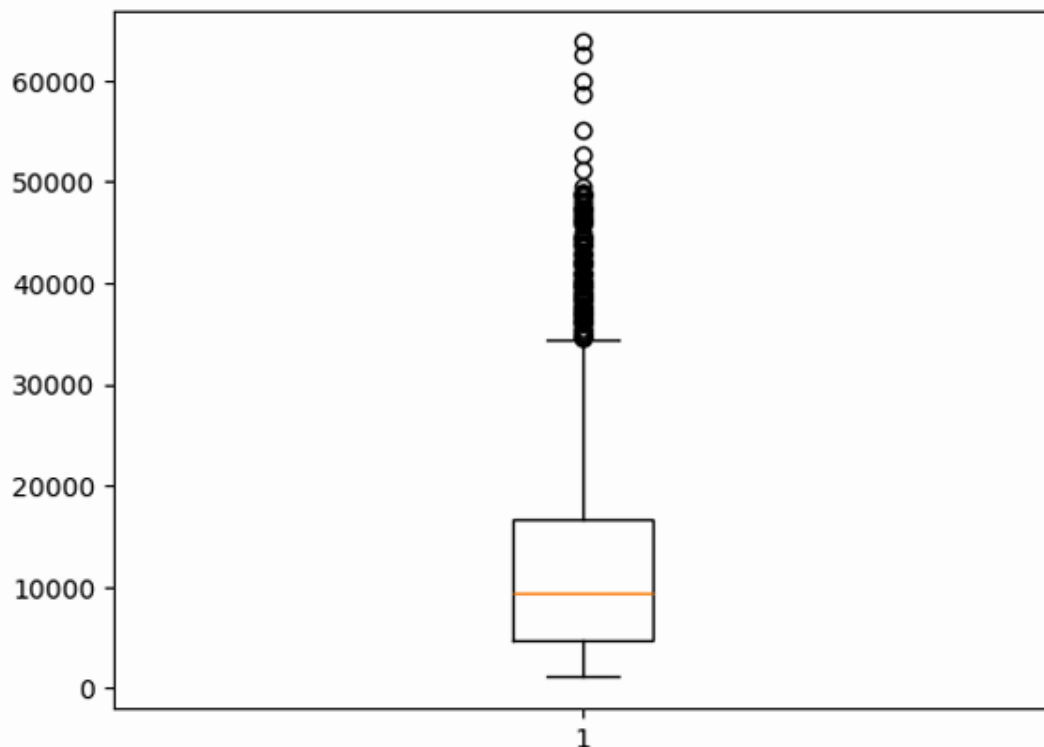
```

import piplite
await piplite.install("seaborn")
np.std(df_num["charges"])
np.float64(12105.484975561612)
import seaborn as sns
mean=np.mean(df_num["charges"])
median=np.median(df_num["charges"])
print(mean)
print(median)
sns.histplot(df_num["charges"],kde=True)
plt.axvline(mean,color="red",linestyle="-",label=f"mean={mean:2f}") \
plt.axvline(median,color="green",linestyle="--",label=f"median={median:2f}")

```



```
df_num["charges"].skew()  
np.float64(1.5158796580240388)  
plt.boxplot(df_num["charges"],vert="false")  
ply.show()
```



```

q1=np.percentile(df_num["charges"],25)
q2=np.percentile(df_num["charges"],50)
q3=np.percentile(df_num["charges"],75) q1,q2,q3
(np.float64(4740.28715), np.float64(9382.033), np.float64(16639.912515))
IQR=q3-q1
IQR
np.float64(11899.625365)
lb=q1-1.5*IQR
ub=q3+1.5*IQR
lb,ub
(np.float64(-13109.1508975), np.float64(34489.350562499996))
conA=df_num["charges"]<=lb
conB=df_num["charges"]<=ub
print(len(df_num[conA | conB]["charges"]))
print(df_num[conA | conB]["charges"])

```

```

1199
0      16884.92400
1      1725.55230
2      4449.46200
3      21984.47061
4      3866.85520
...
1333    10600.54830
1334     2205.98080
1335     1629.83350
1336     2007.94500
1337    29141.36030
Name: charges, Length: 1199, dtype: float64

```

```

df_num_non_outlier=[]
for i in df_num["charges"]:
if i in list(df_num[conA | conB]["charges"]):
df_num_non_outlier.append(mean)
else:
df_num_non_outlier.append(i)
df_num_non_outlier=pd.Series(df_num_non_outlier)
df_num_non_outlier

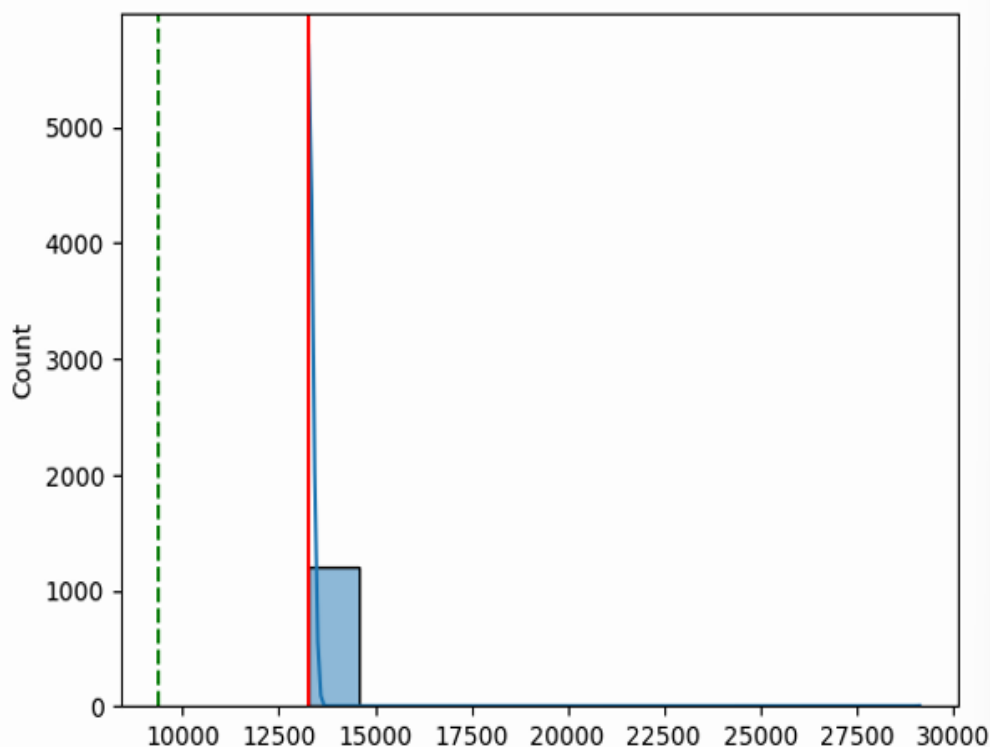
```

```

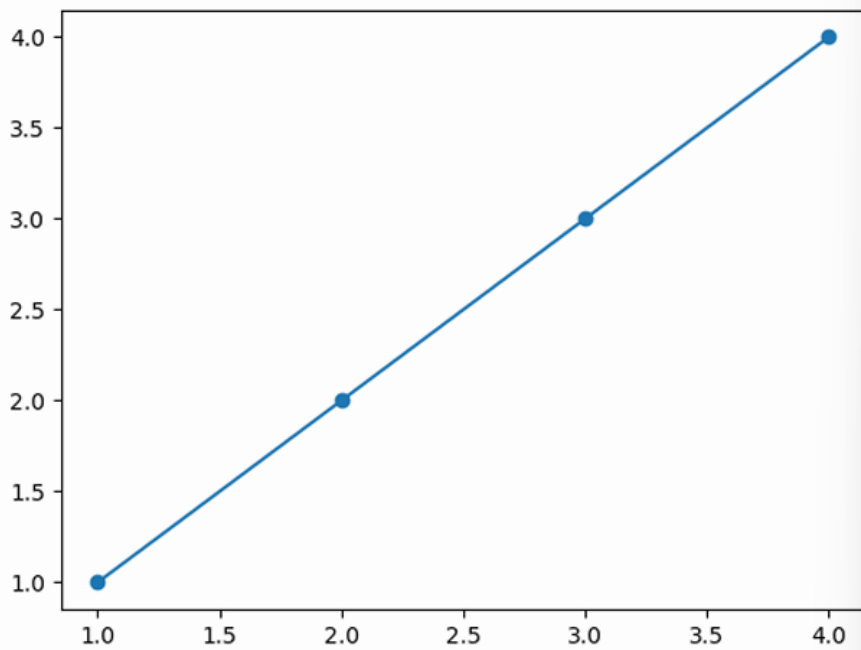
0      13270.422265
1      13270.422265
2      13270.422265
3      13270.422265
4      13270.422265
...
1195    13270.422265
1196    13270.422265
1197    13270.422265
1198    13270.422265
1199    29141.360300
Length: 1200, dtype: float64

```

```
import seaborn as sns
mean=np.mean(df_num_non_outlier)
media=np.median(df_num_non_outlier)
print(mean)
print(media)
sns.histplot(df_num_non_outlier,kde=True)
plt.axvline(mean,color="red",linestyle='-',label=f'mean={mean:2f}')
plt.axvline(media,color="green",linestyle='-',label=f'median={media:2f}')
13283.648046836972 9382.033
```



```
x=list(range(1,5))
y=x plt.scatter(x,y)
plt.plot(x,y)
```



```
df_num.corr()  
sns.heatmap(df_num.corr())
```

